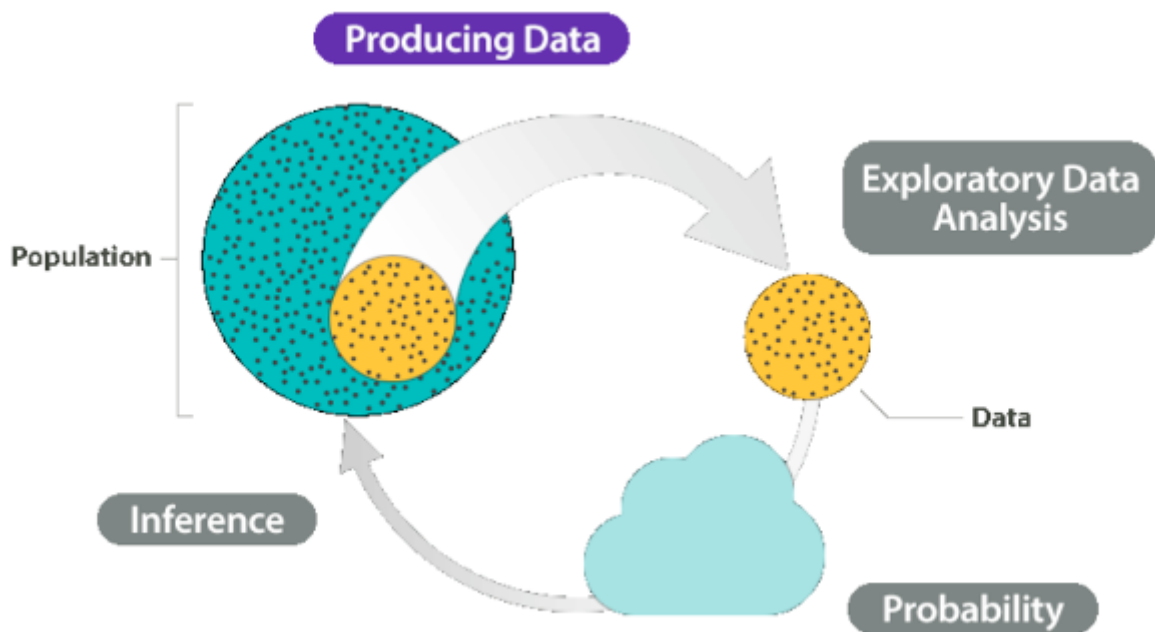


Module 4.4: Statistical Reasoning - Key Terms & Definitions, PART 4

The Big Picture of Statistics: is the four-step process that encompasses statistics: data production, exploratory data analysis, probability, and inference.



Biased Sample: A **sample** that produces data that is not representative because of the systematic under- or over-estimation of the values of the variable of interest is called **biased**. Bias may result from either a poor sampling plan or from a poor design for evaluating the variable of interest.

- **Bias** may arise, for example, when the **sampling frame** does not match the

population of interest. For example, the population of interest consists of all students at the university, whereas the sampling frame consists only of your classmates.

- **Bias is almost guaranteed** if a **volunteer sample** is used (i.e., when individuals select themselves for the study).

Boxplot: The **boxplot** graphically represents the **distribution** of a **quantitative** variable by visually displaying the **five-number summary (min, Q1, M, Q3, Max)** and any observation that was classified as a suspected **outlier** using the **1.5 x (IQR)** criterion.

- There are several ways to plot the whiskers on a boxplot. One convention is to plot whiskers down to the **minimum** and up to the **maximum**. We use the **1.5 x (IQR)** for plotting whiskers.
 1. Calculate the **IQR (Q3 - Q1)**
 2. Multiply the **IQR** by **5**
 3. Add this value to the 75th percentile (**Q3**).
 4. If the value is greater than (or equal to) the maximum value in the dataset, draw the upper whisker to the maximum value. Otherwise, stop the whisker at the largest value that is less than 75th percentile (**Q3**) + **1.5 x IQR**. Plot any values that are greater than this as individual points that are outliers.
 5. Similar, subtract **1.5 x IQR** from the 25th percentile (**Q1**).
 6. If this value is smaller than the minimum value in the dataset, draw the lower whisker to the minimum value. Otherwise, stop the whisker at the lowest value that is greater than 25th percentile (**Q1**) - **1.5 x IQR**. Plot any values that are smaller than this as individual points that are outliers.

Cluster Sampling: This sampling technique is used when our **population** is naturally divided into groups (which we call clusters). For example, all the students in a university are divided into majors. In **cluster sampling**, we take a random sample of clusters, and *use all the individuals within the selected clusters as our sample.*

- Example: In order to get a **sample** of high-school seniors from a certain city, you choose 3 high schools at random from among all the high schools in that city, and use all the high school seniors in the three selected high schools as your **sample**.
- **Cluster vs. Stratified Sampling:** We want to conduct a study on the sleeping habits of UG students at HU, and need to obtain a sample.
 - **Cluster Sampling:** We would randomly choose, say, 2 majors (groups) out of 6, and use all the students in these 2 majors as our sample.
 - **Stratified Sampling:** We would obtain a random sample of, say, 50 students from each of the 6 majors (groups), and use the 300 chosen students as the sample.
 - **Note:** Clearly in this example, **stratified sampling** is much better, since the major of the student might have an effect on the student's sleeping habits, and so we would like to make sure that we have representatives from all the different majors.

Convenience Sample: Individuals were chosen because they were at the right place at the right time to suit the researcher.

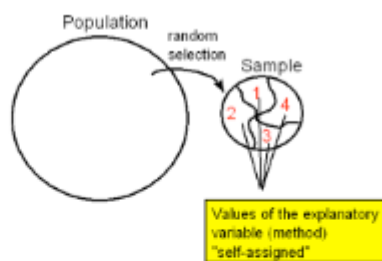
Design Studies:

1. **Observational Study:** The values of the variable or variables of interest are

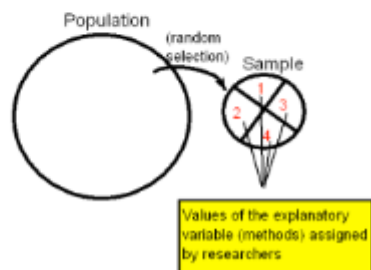
recorded as they naturally occur. There is no inference by the researcher who conducted the study.

2. **Sample Survey:** A particular type of **observational study** in which the individuals report variables' values themselves, frequently by giving their opinions.
3. **Experiment:** Instead of assessing the values of the variables as they naturally occur, the researcher interferes, and they are the ones who assign the values of the explanatory variable of the individuals. The researchers "take control" of the values of the explanatory variable because they want to see how changes in the value of the explanatory variable affect the response variable. (Note: By nature, any experiment involves at least two variables).

1. Observational study:



2. Experiment:



1. *Observational studies:*

- The explanatory variable's values are allowed to occur naturally.
- Because of the possibility of lurking variables, it is difficult to establish causation.
- If possible, control for suspected lurking variables by studying groups of similar individuals.
- Some lurking variables are difficult to control for; others may not be identified.

2. Experiments

- The explanatory variable's values are controlled by researchers (treatment is imposed).
- Randomized assignment to treatments automatically controls for all lurking variables.
- Making subjects blind avoids the placebo effect.
- Making researchers blind avoids conscious or subconscious influences on their subject responses.
- A randomized controlled double-blind experiment is generally optimal for establishing causation.

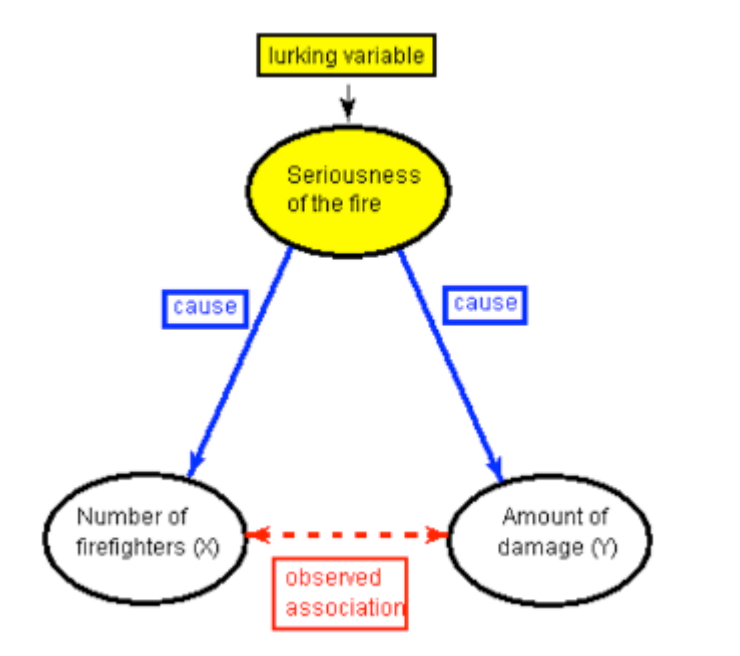
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- A lack of realism may prevent researchers from generalizing experimental results to real-life situations.
 - Noncompliance may undermine an experiment. A volunteer sample might solve (at least partially) this problem.
 - It is impossible, impractical or unethical to impose some treatments.

Law of Large Numbers: The actual (or true) probability of an event (A) is estimated by the relative frequency with which the event occurs in a long series of trials. Put differently, the **law of large numbers** states that as the number of trials increases, the relative frequency becomes the actual probability. So, using this law, as the number of trials increases, the empirical probability gets closer and closer to the theoretical probability.

Lurking Variable: A lurking variable is a variable that is not among the explanatory or response variables in a study, but could substantially affect the interpretation of the

relationship among the variables.

Example:



In this example, the **lurking variable** might have an effect on both the explanatory and the response variable. This *common effect creates the observed association between the explanatory and response variables*, even though there is no causal link between them.

PRINCIPLE: ASSOCIATION DOES NOT IMPLY CAUSATION!

Parameters: A parameter is a number that describes the population; a statistic is a number that is computed from the sample.

- are usually unknown, because it is impractical or impossible to know exactly what values a variable takes for every member of the population.

- Statistics are computed from the sample, and vary from sample due to **sampling variability**.

Numerical Measure	(Population) Parameter	(Sample) Statistic
Proportion	p	$\hat{p}$
Mean	μ	$\bar{x}$
Standard Deviation	σ	s

Population: (Target Group) The whole collection of individuals under the study.

Probability Sampling Plan:

- Cluster Sampling:** This sampling technique is used when our **population** is naturally divided into groups (which we call clusters). For example, all the students in a university are divided into majors. In **cluster sampling**, we take a random sample of clusters, and **use all the individuals within the selected clusters as our sample**.
 - Example: In order to get a sample of high-school seniors from a certain city, you **choose 3 high schools at random from among all the high schools in that city**, and use **all the high school seniors in the three selected high schools as your sample**.
- Stratified Sampling:** When the **population** is naturally divided into sub-populations, which we call stratum, and our sample consists of all these simple random samples put together.
 - Example: In order to get a random sample of high-school seniors from a

certain city, ***we chose of 25 seniors from each of the high schools in that city.***

Our sample consists of all these samples put together.

- **Multistage Sampling:** can have more than 2 stages; makes the sampling process more manageable by working down from a large population to successively smaller groups within the population, taking advantage of **stratifying** along the way, and sometimes finishing up with a **cluster sample** or a **simple random sample**.
 - Example: To obtain a random sample of physicians in the United States, you choose 10 states at random (**stage 1, cluster**). From each state, you choose at random 8 hospitals (**stage 2, cluster**). Finally, from each hospital, you choose 5 physicians from each sub-specialty (**stage 3, stratified**).

Sample: (Sample Member): The observed members of the target group (**population**).

- Note, we should make an effort to choose a sample in such a way that it will represent the population well. For example, if we choose a sample in such a way that it will represent the population of U.S. adults, and ask their opinion about the death penalty, we do not want our sample to consist of only Republicans or only Democrats.
- See also **Biased Sample**

Sample Size:

1. Our first priority is to make sure the sample is representative of the population, by using some form of probability sampling plan.
2. Next, we must keep in mind that in order to get a more precise idea of what values are taken by the variable of interest for the entire population, a larger sample does a better job than a smaller one.
3. Further note: the sample size plays a role in the spread of the distribution of

sample proportion: there should be less spread for larger samples, more spread for smaller samples.

Sample Proportion: notation here is p for population proportion.

Example (AD) Behavior of Sample Proportion $\hat{p}$

Approximately 60% of all part-time college students in the United States are female. (In other words, the population proportion of females among part-time college students is $p = 0.6$.) What would you expect of the behavior of sample proportion of females $\hat{p}$ if random samples of size 100 were taken from the population of all part-time college students?

As we saw before, due to sampling variability, sample proportion in random samples of size 100 will take on many different values which vary according to the laws of chance: in other words, **sample proportion is a random variable**. To summarize the behavior of any random variable, we focus on three features of its distribution: the center, the spread, and the shape.

Based only on our intuition, we would expect the following:

Center: Some sample proportions will be on the low side, say, 0.55 or 0.58 – while others will be on the high side – say, 0.61 or 0.66. It is reasonable to expect all the sample proportions in repeated random samples to average out to the underlying population proportion, 0.6. In other words, the mean of the distribution of sample proportions is 0.6.

Spread: For samples of 100, we would expect sample proportions of females not to stray too far from population proportion 0.6. Sample proportions lower than 0.5 or higher than 0.7 would be rather surprising. On the other hand, if we were only taking samples of size 10, we would not be at all surprised by a sample proportion of females even as low as $4/10 = 0.4$, or as high as $8/10 = 0.8$. Thus, sample size plays a role in the distribution of sample proportion: there should be less spread for larger samples, more spread for smaller samples.

Shape: Sample proportions closest to 0.6 would be most common, and sample proportions far from 0.6 in either direction would be progressively less likely. In other words, the shape of the distribution of sample proportions should bulge in the middle and taper at the ends: it should be somewhat *normal*.

Sampling Distribution The *distribution* of the values of the sample proportions $\hat{p}$ in repeated samples is called the *sampling distribution* of $\hat{p}$.

Watch this Simulation The purpose of this activity is to check whether our intuition about the center and shape of the sampling distribution of $\hat{p}$ was right via simulations.

Watch the simulation here: <https://youtu.be/2bIC4EmejkQ>.

At this point, we have a good sense of what happens as we take random samples from a population. Our intuition suggests that our initial intuition about the shape and center of the sampling distribution is correct. If a population has a proportion of p , then random samples of the same size drawn from the population will have sample proportions close to p . More specifically, the distribution of sample proportions will have a mean of p . We also observed that in this situation, the sample proportions are approximately normal. We will see later that this is not always true: if sample proportions are normally distributed, then the distribution is centered at p .

Sampling Distribution: The distribution of the values of the sample proportions $\hat{p}$ in repeated samples is called the sampling distribution of $\hat{p}$.

The Sampling Distribution of the Sample Proportion If repeated random samples of a given size n are taken from a population of values for a categorical variable, where the proportion in the category of interest is p , then the mean of all sample proportions $\hat{p}$ is the population proportion (p). As for the spread of all sample proportions, theory describes the behavior much more precisely than saying that there is less spread for larger samples. In fact, the standard deviation of all sample proportions $\hat{p}$ is exactly $\sqrt{\frac{p(1-p)}{n}}$.

Since sample size n appears in the denominator of the square root, the standard deviation does decrease as sample size increases. Finally, the shape of the distribution of $\hat{p}$ will be approximately normal as long as the sample size n is large enough. The convention is to require both np and $n(1-p)$ to be at least 10.

- We can summarize all of the above as the following:

$\hat{p}$ has a normal distribution with a mean of $\mu_{\hat{p}} = p$ and standard deviation $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$ (and as long as np and $n(1-p)$ are at least 10).

Let's apply this result to our example and see how it compares with our simulation.

In our example, $n = 25$ (sample size) and $p = 0.6$. Note that $np = 15 \geq 10$ and $n(1-p) = 10 \geq 10$. Therefore we

can conclude that $\hat{p}$ is approximately a normal distribution with mean $p = 0.6$ and standard deviation $\sqrt{\frac{p(1-p)}{n}}$

$$\sqrt{\frac{0.6(1-0.6)}{25}} = 0.097$$

Sampling Distribution of the Sampling Mean:

The Sampling Distribution of the Sample Mean If repeated random samples of a given size n are taken from a population of values for a quantitative variable, where the population mean is μ and the population standard deviation is σ , then the mean of all sample means ($\bar{x}$) is population mean μ . As for the spread of all sample means, theory describes the behavior much more precisely than saying that there is less spread for larger samples. In fact, the standard deviation of all sample means ($\bar{x}$) is exactly $\frac{\sigma}{\sqrt{n}}$. Since the square root of sample size n appears in the denominator, the standard deviation does decrease as sample size increases.

Let’s compare and contrast what we now know about the sampling distributions for sample means and proportions.

			Sampling Distribution		
Variable	Parameter	Statistic	Center	Spread	Shape
Categorical (example: left-handed or not)	p = population proportion	$\hat{p}$ = sample proportion	p	$\sqrt{\frac{p(1-p)}{n}}$	Normal, if $np \geq 10$ and $n(1-p) \geq 10$
Quantitative (example: age)	μ = population mean, σ = population standard deviation	$\bar{x}$ = sample mean	μ	$\frac{\sigma}{\sqrt{n}}$	Normal, if $n \geq 30$

Sampling Distributions for Sample Means and Sample Proportions:

			Sampling Distribution	
Variable	Parameter	Statistic	Center	Spread
Categorical (example: left-handed or not)	p = population proportion	$\hat{p}$ = sample proportion	p	$\sqrt{\frac{p(1-p)}{n}}$
Quantitative (example: age)	μ = population mean, σ = population standard deviation	$\bar{x}$ = sample mean	μ	$\frac{\sigma}{\sqrt{n}}$

- Statistics vary from sample to sample due to **sampling variability**, and therefore can be regarded as **random variables** whose distribution we call the **sampling distribution**.
 - Sample proportion, $\hat{p}$** , and the same **mean, μ** , relative to their respective population parameters (p & μ).
 - Under certain conditions the **sampling distribution** is approximately normal. This result is known as the **Central Limit Theorem** – the foundation for statistical inference.

Sampling Frame: List of potential individuals to be sampled.

Simple Random Sampling: the simplest probability sampling plan. It is equivalent to “selecting names out of a hat.” Each individual has the same chance of being selected.

Simpson’s Paradox: Whenever a **lurking variable** causes us to rethink the direction of association, we call it the **Simpson’s paradox**.

Stratified Sampling: When the **population** is naturally divided into sub-populations, which we call stratum, and our sample consists of all these simple random samples put together.

- Example: In order to get a random sample of high-school seniors from a certain city, ***we choose of 25 seniors from each of the high schools in that city.*** Our sample

consists of all these samples put together.

- **Cluster vs. Stratified Sampling:** We want to conduct a study on the sleeping habits of UG students at HU, and need to obtain a sample.
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 - **Note:** Clearly in this example, **stratified sampling** is much better, since the major of the student might have an effect on the student's sleeping habits, and so we would like to make sure that we have representatives from all the different majors.

Systematic Sampling: takes an organized (but not random) approach to the selection process, as in picking every 50th name on a list, or the first product to come off the production line each hour.

Variable: A particular characteristic of the *individual*.

- **Variables** can be classified into *one of two types*:
 - **Quantitative variables:** take numerical values and represent some kind of measurement (e.g., weight, age, height, etc.)
 - **Categorical variables:** take category or label values and place an individual into one of several groups. Each observation can be placed in only one category, and the categories are mutually exclusive (e.g., zipcode, gender, race, smoking yes/no, etc.)
 - The **distribution** of a **categorical variable** is summarized using a *graphical display*: a **pie chart** or a **bar chart**, supplemented by

numerical summaries: category counts and percentages.

Sample Surveys: are occasionally used to examine relationships, but often they assess values of many separate variables, such as respondents' opinions on various matters. Survey questions should be designed carefully, in order to ensure an **unbiased** assessment of the variables' values.